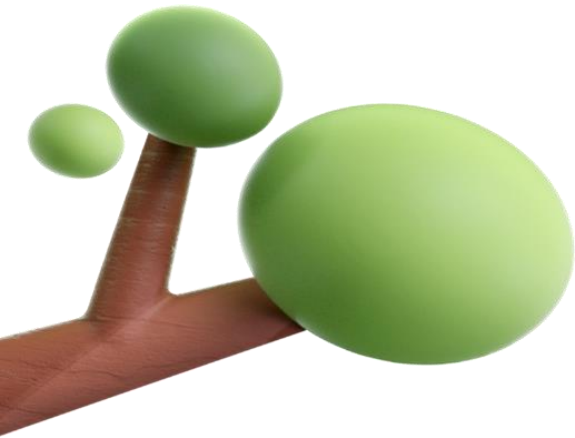




LEVEL 3 ASSIGNMENT

STEP PROGRAM

STRINGS

SUBMITTED BY:

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SOURCE CODE

LEVEL 3 QUESTION: 1

Q. Write a program to find and return the length of a string without using the length() method

```
import java.util.Scanner;
```

```
public class StringLengthWithoutLengthMethod {
```

```
    public static int findStringLength(String text) {
```

```
        int length = 0;
```

```
        try {
```

```
            while (true) {
```

```
                text.charAt(length);
```

```
                length++;
```

```
            }
```

```
        } catch (IndexOutOfBoundsException e) {
```

```
        }
```

```
        return length;
```

```
    }
```

```
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

System.out.print("Enter a string: ");

String input = sc.next();

int lengthWithoutMethod = findStringLength(input);

int lengthWithMethod = input.length();

System.out.println("Length of the string using user-defined method: " +
lengthWithoutMethod);

System.out.println("Length of the string using built-in length() method: " +
lengthWithMethod);

}

}
```

OUTPUT:

```
Enter a string: hello
Length of the string using user-defined method: 5
Length of the string using built-in length() method: 5
```

LEVEL 3 QUESTION: 2

Q. Write a program to split the text into words, compare the result with the split() method and

display

the

result

```
import java.util.Scanner;
```

```
public class SplitTextComparison {
```

```
    public static int findStringLength(String text) {
```

```
        int length = 0;
```

```
        try {
```

```
            while (true) {
```

```
                text.charAt(length);
```

```
                length++;
```

```
            }
```

```
        } catch (IndexOutOfBoundsException e) {
```

```
        }
```

```
        return length;
```

```
    }
```

```
    public static String[] splitTextManually(String text) {
```

```
        int length = findStringLength(text);
```

```
        int wordCount = 0;
```

```
        for (int i = 0; i < length; i++) {
```

```

        if (text.charAt(i) == ' ') {

            wordCount++;

        }
    }

    wordCount++;

    String[] words = new String[wordCount];

    int startIndex = 0;

    int wordIndex = 0;

    for (int i = 0; i < length; i++) {

        if (text.charAt(i) == ' ' || i == length - 1) {

            words[wordIndex++] = text.substring(startIndex, i == length
- 1 ? i + 1 : i);

            startIndex = i + 1;

        }

    }

    return words;

}

public static boolean compareArrays(String[] array1, String[]
array2) {

    if (array1.length != array2.length) {

```

```
        return false;
    }

    for (int i = 0; i < array1.length; i++) {
        if (!array1[i].equals(array2[i])) {
            return false;
        }
    }

    return true;
}
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter a text: ");
    String input = sc.nextLine();

    String[] manualSplit = splitTextManually(input);
    String[] builtInSplit = input.split(" ");

    boolean isEqual = compareArrays(manualSplit, builtInSplit);

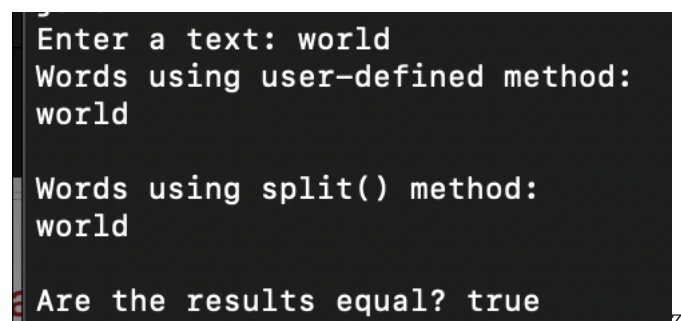
    System.out.println("Words using user-defined method:");
    for (String word : manualSplit) {
```

```
        System.out.println(word);
    }

    System.out.println("\nWords using split() method:");
    for (String word : builtInSplit) {
        System.out.println(word);
    }

    System.out.println("\nAre the results equal? " + isEqual);
}
}
```

OUTPUT:

A screenshot of a Java program's output in a terminal window. The output shows the text 'world' being entered, followed by 'Words using user-defined method: world', 'Words using split() method: world', and 'Are the results equal? true'.

```
Enter a text: world
Words using user-defined method:
world

Words using split() method:
world

Are the results equal? true
```

LEVEL 3 QUESTION: 3

Q. Write a program to split the text into words and return the words along with their lengths in a 2D array

```
import java.util.Scanner;

public class TextSplitWithLengths {

    public static int findStringLength(String text) {

        int length = 0;

        try {

            while (true) {

                text.charAt(length);

                length++;

            }

        } catch (IndexOutOfBoundsException e) {

        }

        return length;

    }

    public static String[] splitTextManually(String text) {

        int length = findStringLength(text);

        int wordCount = 0;

        for (int i = 0; i < length; i++) {

            if (text.charAt(i) == ' ') {
```

```
        wordCount++;

    }

}

wordCount++;

String[] words = new String[wordCount];

int startIndex = 0;

int wordIndex = 0;

for (int i = 0; i < length; i++) {

    if (text.charAt(i) == ' ' || i == length - 1) {

        words[wordIndex++] = text.substring(startIndex, i == length - 1 ? i + 1 : i);

        startIndex = i + 1;

    }

}

return words;

}
```

```
public static String[][] getWordsWithLengths(String[] words) {

    String[][] result = new String[words.length][2];

    for (int i = 0; i < words.length; i++) {

        result[i][0] = words[i];
```

```

        result[i][1] = String.valueOf(words[i].length());
    }

    return result;
}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a text: ");

    String input = sc.nextLine();

    String[] words = splitTextManually(input);

    String[][] wordsWithLengths = getWordsWithLengths(words);

    System.out.println("Word\tLength");

    for (int i = 0; i < wordsWithLengths.length; i++) {

        System.out.println(wordsWithLengths[i][0] + "\t" +
Integer.parseInt(wordsWithLengths[i][1]));

    }

}
}

```

OUTPUT:

```
.java
Enter a text: hello
Word      Length
hello     5
```

LEVEL 3 QUESTION: 4

Q. Write a program to split the text into words and find the shortest and longest strings in a given text

```
import java.util.Scanner;
```

```
public class TextAnalysis {
```

```
    public static int findStringLength(String text) {
```

```
        int length = 0;
```

```
        try {
```

```
            while (true) {
```

```
                text.charAt(length);
```

```
                length++;
```

```
            }
```

```
        } catch (IndexOutOfBoundsException e) {
```

```
        }
```

```
        return length;
```

```
    }
```

```
public static String[] splitTextManually(String text) {

    int length = findStringLength(text);

    int wordCount = 0;

    for (int i = 0; i < length; i++) {

        if (text.charAt(i) == ' ') {

            wordCount++;

        }

    }

    wordCount++;

    String[] words = new String[wordCount];

    int startIndex = 0;

    int wordIndex = 0;

    for (int i = 0; i < length; i++) {

        if (text.charAt(i) == ' ' || i == length - 1) {

            words[wordIndex++] = text.substring(startIndex, i == length - 1 ? i + 1 : i);

            startIndex = i + 1;

        }

    }

    return words;
}
```

```
}
```

```
public static String[][] getWordsWithLengths(String[] words) {
```

```
    String[][] result = new String[words.length][2];
```

```
    for (int i = 0; i < words.length; i++) {
```

```
        result[i][0] = words[i];
```

```
        result[i][1] = String.valueOf(words[i].length());
```

```
    }
```

```
    return result;
```

```
}
```

```
public static int[] findShortestAndLongest(String[][] wordsWithLengths) {
```

```
    int shortestLength = Integer.MAX_VALUE;
```

```
    int longestLength = Integer.MIN_VALUE;
```

```
    int shortestIndex = -1;
```

```
    int longestIndex = -1;
```

```
    for (int i = 0; i < wordsWithLengths.length; i++) {
```

```
        int wordLength = Integer.parseInt(wordsWithLengths[i][1]);
```

```
        if (wordLength < shortestLength) {
```

```
            shortestLength = wordLength;
```

```
        shortestIndex = i;
    }

    if (wordLength > longestLength) {

        longestLength = wordLength;

        longestIndex = i;
    }
}

return new int[]{shortestIndex, longestIndex};
}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a text: ");

    String input = sc.nextLine();

    String[] words = splitTextManually(input);

    String[][] wordsWithLengths = getWordsWithLengths(words);

    int[] shortestAndLongest = findShortestAndLongest(wordsWithLengths);

    System.out.println("Shortest Word: " + words[shortestAndLongest[0]]);

    System.out.println("Longest Word: " + words[shortestAndLongest[1]]);
}
```

```
}  
  
}
```

OUTPUT:

```
Enter a text: car  
Shortest Word: car  
Longest Word: car
```

LEVEL 3 QUESTION: 5

Q. Write a program to find vowels and consonants in a string and display the count of Vowels and Consonants in the string

```
import java.util.Scanner;
```

```
public class VowelConsonantCounter {
```

```
    public static String checkCharacterType(char ch) {
```

```
        ch = Character.toLowerCase(ch);
```

```
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
```

```
            return "Vowel";
```

```
        } else if (ch >= 'a' && ch <= 'z') {
```

```
            return "Consonant";
```

```
    } else {  
        return "Not a Letter";  
    }  
}  
  
public static int[] countVowelsAndConsonants(String text) {  
    int vowels = 0;  
    int consonants = 0;  
  
    for (int i = 0; i < text.length(); i++) {  
        char ch = text.charAt(i);  
  
        if (checkCharacterType(ch).equals("Vowel")) {  
            vowels++;  
        } else if (checkCharacterType(ch).equals("Consonant")) {  
            consonants++;  
        }  
    }  
  
    return new int[] {vowels, consonants};  
}  
  
public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

System.out.print("Enter a text: ");

String input = sc.nextLine();


int[] result = countVowelsAndConsonants(input);


System.out.println("Vowels Count: " + result[0]);

System.out.println("Consonants Count: " + result[1]);

}

}
```

OUTPUT:

```
1.java
Enter a text: class
Vowels Count: 1
Consonants Count: 4
(base) harikrishna@Harikrishna
```

LEVEL 3 QUESTION: 6

Q. Write a program to find vowels and consonants in a string and display the character type - Vowel, Consonant, or Not a Letter

```
import java.util.Scanner;
```

```
public class CharacterTypeChecker {
```

```
    public static String checkCharacterType(char ch) {
```

```
        ch = Character.toLowerCase(ch);
```

```
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
```

```
            return "Vowel";
```

```
        } else if (ch >= 'a' && ch <= 'z') {
```

```
            return "Consonant";
```

```
        } else {
```

```
            return "Not a Letter";
```

```
        }
```

```
    }
```

```
    public static String[][] findCharacterTypes(String text) {
```

```
        String[][] result = new String[text.length()][2];
```

```

        for (int i = 0; i < text.length(); i++) {

            char ch = text.charAt(i);

            result[i][0] = String.valueOf(ch);

            result[i][1] = checkCharacterType(ch);

        }

        return result;

    }

    public static void displayCharacterTypes(String[][] characterTypes) {

        System.out.println("Character\tType");

        for (int i = 0; i < characterTypes.length; i++) {

            System.out.printf("%-10s\t%-10s\n",                characterTypes[i][0],
characterTypes[i][1]);

        }

    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a text: ");

        String input = sc.nextLine();

        String[][] result = findCharacterTypes(input);

```

```
        displayCharacterTypes(result);  
    }  
}
```

OUTPUT:

```
Enter a text: hello  
Character      Type  
h              Consonant  
e              Vowel  
l              Consonant  
l              Consonant  
o              Vowel
```

LEVEL 3 QUESTION: 7

Q. Write a program to trim the leading and trailing spaces from a string using the charAt() method

```
import java.util.Scanner;  
  
public class TrimStringUsingCharAt {  
  
    public static int[] trimSpaces(String text) {  
  
        int start = 0;  
  
        int end = text.length() - 1;
```

```
while (start <= end && text.charAt(start) == ' ') {

    start++;

}

while (end >= start && text.charAt(end) == ' ') {

    end--;

}

return new int[]{start, end};

}

public static String getSubstring(String text, int start, int end) {

    StringBuilder result = new StringBuilder();

    for (int i = start; i <= end; i++) {

        result.append(text.charAt(i));

    }

    return result.toString();

}

public static boolean compareStrings(String str1, String str2) {

    if (str1.length() != str2.length()) {

        return false;

    }

}
```

```

    }

    for (int i = 0; i < str1.length(); i++) {

        if (str1.charAt(i) != str2.charAt(i)) {

            return false;

        }

    }

    return true;

}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a string with leading and trailing spaces: ");

    String input = sc.nextLine();

    int[] trimmedIndexes = trimSpaces(input);

    String  trimmedUsingCharAt  =  getSubstring(input,  trimmedIndexes[0],
trimmedIndexes[1]);

    String trimmedUsingBuiltIn = input.trim();

    System.out.println("Trimmed using charAt() method: '" + trimmedUsingCharAt +
    "'");

```

```

        System.out.println("Trimmed using built-in trim() method: " +
trimmedUsingBuiltIn + "");

        boolean areEqual = compareStrings(trimmedUsingCharAt,
trimmedUsingBuiltIn);

        System.out.println("Both methods give the same result: " + areEqual);
    }
}

```

OUTPUT:

```

C:\java>
Enter a string with leading and trailing spaces: TrimStringUsingCharAt
Trimmed using charAt() method: 'TrimStringUsingCharAt'
Trimmed using built-in trim() method: 'TrimStringUsingCharAt'
Both methods give the same result: true

```

LEVEL 3 QUESTION: 8

Q. Write a program to take user input for the age of all 10 students in a class and check whether the student can vote depending on his/her age is greater or equal to 18.

```

import java.util.Random;

import java.util.Scanner;

```

```
public class VotingEligibility {

    public static int[] getRandomAges(int n) {

        Random random = new Random();

        int[] ages = new int[n];

        for (int i = 0; i < n; i++) {

            ages[i] = 18 + random.nextInt(83);

        }

        return ages;

    }

    public static String[][] checkVotingEligibility(int[] ages) {

        String[][] result = new String[ages.length][2];

        for (int i = 0; i < ages.length; i++) {

            if (ages[i] < 0) {

                result[i][0] = String.valueOf(ages[i]);

                result[i][1] = "Cannot Vote (Invalid Age)";

            } else if (ages[i] >= 18) {
```

```
        result[i][0] = String.valueOf(ages[i]);

        result[i][1] = "Can Vote";

    } else {

        result[i][0] = String.valueOf(ages[i]);

        result[i][1] = "Cannot Vote";

    }

}

return result;

}

public static void displayResult(String[][] result) {

    System.out.println("Age\tCan Vote?");

    System.out.println("-----");

    for (int i = 0; i < result.length; i++) {

        System.out.println(result[i][0] + "\t" + result[i][1]);

    }

}
```

```
public static void main(String[] args) {  
  
    Scanner sc = new Scanner(System.in);  
  
    System.out.print("Enter the number of students (e.g. 10): ");  
  
    int n = sc.nextInt();  
  
    int[] ages = getRandomAges(n);  
  
    String[][] eligibility = checkVotingEligibility(ages);  
  
    displayResult(eligibility);  
}  
}
```

OUTPUT:

```
Enter the number of students (e.g. 10): 7
Age      Can Vote?
-----
27       Can Vote
30       Can Vote
53       Can Vote
70       Can Vote
37       Can Vote
78       Can Vote
20       Can Vote
```

LEVEL 3 QUESTION: 9

Q. Rock-Paper-Scissors is a game played between a minimum of two players. Each player can choose either rock, paper, or scissors. Here the game is played between a user and a computer. Based on the rules, either a player or a computer will win. Show the stats of player and computer win in a tabular format across multiple games. Also, show the winning percentage between the player and the computer.

```
import java.util.Scanner;
```

```
public class RockPaperScissors {
```

```
    public static String getComputerChoice() {
```

```
        int randomChoice = (int) (Math.random() * 3);
```

```
        if (randomChoice == 0) return "rock";
```

```
        else if (randomChoice == 1) return "paper";
```

```
        else return "scissors";
```

```
}
```

```
public static String findWinner(String userChoice, String  
computerChoice) {
```

```
    if (userChoice.equals(computerChoice)) return "Draw";
```

```
    if ((userChoice.equals("rock") &&  
computerChoice.equals("scissors")) ||
```

```
        (userChoice.equals("paper") &&  
computerChoice.equals("rock")) ||
```

```
        (userChoice.equals("scissors") &&  
computerChoice.equals("paper")))) {
```

```
        return "Player";
```

```
    }
```

```
    return "Computer";
```

```
}
```

```
public static String[][] calculateStats(int playerWins, int  
computerWins, int gamesPlayed) {
```

```
    double playerWinPercentage = (double) playerWins /  
gamesPlayed * 100;
```

```
double    computerWinPercentage    =    (double)
computerWins / gamesPlayed * 100;
```

```
String[][] stats = new String[gamesPlayed + 1][4];
```

```
for (int i = 0; i < gamesPlayed; i++) {

    stats[i][0] = "Game " + (i + 1);

    stats[i][1] = playerWins > computerWins ? "Player
Wins" : "Computer Wins";

    stats[i][2] = String.valueOf(playerWins);

    stats[i][3] = String.valueOf(computerWins);

}
```

```
stats[gamesPlayed][0] = "Total Wins";

stats[gamesPlayed][1] = "Player: " + playerWins + ",
Computer: " + computerWins;

stats[gamesPlayed][2]  =  String.format("Player  Win
Percentage: %.2f%%", playerWinPercentage);

stats[gamesPlayed][3] = String.format("Computer Win
Percentage: %.2f%%", computerWinPercentage);
```

```
        return stats;
    }

    public static void displayStats(String[][] stats) {

        System.out.println("Game Results:");

        System.out.println("Game\tPlayer\tComputer");

        for (int i = 0; i < stats.length - 1; i++) {

            System.out.println(stats[i][0] + "\t" + stats[i][1] + "\t"
+ stats[i][2]);

        }

        System.out.println("Total Stats:");

        System.out.println(stats[stats.length - 1][0]);

        System.out.println(stats[stats.length - 1][1]);

        System.out.println(stats[stats.length - 1][2]);

        System.out.println(stats[stats.length - 1][3]);

    }

    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

System.out.print("Enter number of games: ");

int numberOfGames = sc.nextInt();

int playerWins = 0;

int computerWins = 0;

for (int i = 0; i < numberOfGames; i++) {

    System.out.print("Enter your choice (rock, paper, or
scissors): ");

    String userChoice = sc.next().toLowerCase();

    String computerChoice = getComputerChoice();

    System.out.println("Computer's choice: " +
computerChoice);

    String winner = findWinner(userChoice,
computerChoice);

    if (winner.equals("Player")) playerWins++;
```

```
        else if (winner.equals("Computer"))
computerWins++;

        System.out.println("Winner: " + winner);
    }

    String[][] stats = calculateStats(playerWins,
computerWins, numberOfGames);

    displayStats(stats);
}
}
```

OUTPUT:

```
Enter number of games: 2
Enter your choice (rock, paper, or scissors): rock
Computer's choice: paper
Winner: Computer
Enter your choice (rock, paper, or scissors): paper
Computer's choice: scissors
Winner: Computer
Game Results:
Game      Player  Computer
Game 1    Computer Wins    0
Game 2    Computer Wins    0
Total Stats:
Total Wins
Player: 0, Computer: 2
Player Win Percentage: 0.00%
Computer Win Percentage: 100.00%
(base) harikrishna@Harikrishnas-MacBook-Air-281 java %
```

LEVEL 3 QUESTION: 10

Q. Create a program to take input marks of students in 3 subjects physics, chemistry, and maths. Compute the percentage and then calculate the grade as shown in figure below

```
import java.util.*;
```

```
public class StudentScoreCard {
```

```
    public static int[][] generatePCMScores(int students) {
```

```
        int[][] scores = new int[students][3];
```

```
        for (int i = 0; i < students; i++) {
```

```
        for (int j = 0; j < 3; j++) {  
            scores[i][j] = 40 + (int) (Math.random() * 61); // scores  
between 40 and 100  
        }  
    }  
  
    return scores;  
}
```

```
public static double[][] calculateScoreStats(int[][] scores) {  
    double[][] stats = new double[scores.length][3];  
  
    for (int i = 0; i < scores.length; i++) {  
        int total = scores[i][0] + scores[i][1] + scores[i][2];  
  
        double average = total / 3.0;  
  
        double percentage = (total / 300.0) * 100;  
  
        stats[i][0] = total;  
  
        stats[i][1] = Math.round(average * 100.0) / 100.0;  
  
        stats[i][2] = Math.round(percentage * 100.0) / 100.0;  
    }  
}
```

```
        return stats;
    }

    public static String[] assignGrades(double[][] stats) {
        String[] grades = new String[stats.length];
        for (int i = 0; i < stats.length; i++) {
            double percentage = stats[i][2];
            if (percentage >= 90) grades[i] = "A+";
            else if (percentage >= 80) grades[i] = "A";
            else if (percentage >= 70) grades[i] = "B+";
            else if (percentage >= 60) grades[i] = "B";
            else if (percentage >= 50) grades[i] = "C";
            else if (percentage >= 40) grades[i] = "D";
            else grades[i] = "F";
        }
        return grades;
    }
}
```

```
public static void displayScoreCard(int[][] scores, double[][] stats, String[] grades) {
```

```
    System.out.printf("%-8s %-8s %-8s %-8s %-8s %-8s %-10s %-6s\n", "Student", "Physics", "Chemistry", "Maths", "Total", "Average", "Percentage", "Grade");
```

```
    for (int i = 0; i < scores.length; i++) {
```

```
        System.out.printf("%-8s %-8d %-8d %-8d %-8.0f %-8.2f %-10.2f %-6s\n",
```

```
            "Student" + (i + 1), scores[i][0], scores[i][1], scores[i][2],
```

```
            stats[i][0], stats[i][1], stats[i][2], grades[i]);
```

```
    }
```

```
}
```

```
public static void main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    System.out.print("Enter number of students: ");
```

```
    int n = sc.nextInt();
```

```
    int[][] scores = generatePCMScores(n);
```

```
    double[][] stats = calculateScoreStats(scores);
```

```
String[] grades = assignGrades(stats);

displayScoreCard(scores, stats, grades);

}

}
```

OUTPUT:

```
Enter number of students: 2
Student Physics Chemistry Maths Total Average Percentage Grade
Student1 53 89 44 186 62.00 62.00 B
Student2 43 82 80 205 68.33 68.33 B
(base) harikrishna@Harikrishnas-MacBook-Air-281: java %
```